

Course Code	21CSE597T	Course Name	CONTAINERS AND CLOUD DevOps		Course Category	E	PROFESSIONAL ELECTIVE			L	T	P	C
										3	0	0	3
Pre-requisite Courses	Nil		Co-requisite Courses	Nil		Progressive Courses	Nil						
Course Offering Department	School of Computing		Data Book / Codes/Standards		Nil								
Course Learning Rationale (CLR):													
The purpose of learning this course is to:													
CLR-1:	learn and understand how containers impact the development and deployment of software applications												
CLR-2:	understand why DevOps is needed and has become indispensable to today's infrastructure												
CLR-3:	modern applications follow an agile process thus requiring frequent updates to be pushed out												
CLR-4:	predict of deployments and application bundling is key to reduce post deployment failures												
CLR-5:	infrastructure as code is core to repeatable infrastructure creation and tear down Container orchestration allows customers to scale applications very effectively across platforms												
Course Outcomes (CO):													
At the end of this course, learners will be able to:													
CO-1:	Install, configure a Docker container system												
CO-2:	Create a DevOps plan, choose the right tools and implement each phase with CI/CD												
CO-3:	Apply principles of code version management												
CO-4:	Automate build push to IaaS												
CO-5:	Apply cloud automation techniques												
Module 1: Introduction													
Introduction to Docker Ecosystem, DevOps Toolchain, Git Essentials													
Module 2: AWS													
Introduction to AWS, AWS Entities, AWS container entities , AWS ECR, AWS ECS, AWS Code Commit, AWS Code Build, Code Deploy, Code Pipeline, Jenkins													
Module 3: IaC and Configuration management													
Cloud Formation, Terraform, Ansible, Infrastructure automation for Multi service deployment through CloudFormation, workflow management with Terraform, Ansible & Automating management of cloud services													
Module 4 : Observability													
IAM, CloudTrail, CloudWatch, CLI, Lambda, Kinesis, S3, IAM (Identity and Access Management), Event Logging and Insights using CloudTrail, Visualizing using CloudWatch , CLI (Command Line Interface), Lambda, Creating, deploying, and managing Lambda functions.													
Module 5: Containerized workload management													
Container Orchestration cluster Introduction to Elastic Kubernetes Service (EKS), Azure Kubernetes service, Containers and multi cloud deployment strategies													